

Tools that are part of gEDA

Tool	Usage
gschem	Schematic capture
gnetlist	Netlist generator
gattrib	Attribute editor
symbols	Symbol library
libgeda	Libraries for gschem gnetlist and gsymcheck
gsymcheck	Symbol checker
examples	Example projects
documentation	Documentation

Table 1

fabrication, the following tools are used:

- *gschem* – draws electrical schematic
- *gsch2pcb/gnetlist* – creates netlist
- *PCB* – reads netlist, draws PCB
- *gerbv* – Gerber checker/viewer

The gEDA project is widely aimed at circuits of low- to mid-level complexity that are geared towards board-level designs.

Why gEDA?

The first thing that makes it so attractive is its GPL license. gEDA supports a wide range of platforms including GNU/Linux, UNIX, OSX and Windows. Besides these, the following are some of the main attractions:

- File formats are in ASCII text and well documented.
- Command line options help to automate design flows using UNIX tools like make, awk, sed, etc.
- There are no boundaries for design size, number of components, PCB layer count, etc.
- Free from DRM—i.e., you can share your designs and design elements with everyone.

Last, but not the least, is the superb mailing list of the project, where you can find developments, bug fixes and user queries being discussed.

On the downside, there might be some cutting-edge features that you find missing from gEDA. However, the features available are adequate, and more are being added over time.

Component and symbol library

A good EDA tool requires a wide collection of symbols, which are a simple representation of circuit elements such as resistors, ICs, connectors, etc. This is available in the gEDA library—currently, it numbers approximately 2,000. All these symbols are in ASCII formats so they can be easily edited using text editors. Note that the schematic is a collection of symbols, and thus all schematics have the same format as symbols.

Symbol libraries are always never complete as new components are developed over time. So regular updating of the symbol library is vital. An online collection of symbols is maintained at gedasymbols.org. It's a repository for symbols, footprints and documentation that is useful to

Separately developed/maintained tools, but associated with gEDA

Tool	Usage
PCB	Free/open source PCB layout editor
gerbv	Gerber viewer
gnucap	GPL'ed mixed-mode/mixed-level circuit simulator
gnucap+	An extension of the GNU Circuit Analysis package
GTK Wave	Electronic waveform viewer
gwave	Analogue waveform viewer
Icarus Verilog	Verilog simulation and synthesis tool
ngspice	An improved SPICE
gsch2pcb	A tool to forward annotation from your schematic to layout, using pcb
GSpiceUI	Graphical front-end to SPICE/gnucap
wcalc	Transmission line analyser
vhdl2vl	VHDL to Verilog translator

Table 2

anyone using *gschem* and *pcb*. The mission of gedasymbols.org is to increase the usage of the gEDA suite. Users who would like to contribute missing symbols can add them to the online library. The updated components/symbols library will be available during its next release.

As with all EDA tools, *gschem* has a symbol editor. You can draw your symbol and save it. You can upload this to gedasymbols.org.

Draw schematics with gschem

gschem is the specialised drawing program for gEDA. The term *gschem* means gEDA's schematic. Usually you'll need

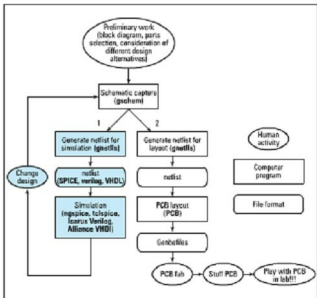


Figure 1: Board design flow using gEDA